

A Z-Curve Credibility Audit of Face Selectivity in the Fusiform Face Area (#276068)

Author(s)

This pre-registration is currently anonymous to enable blind peer-review.
It has 5 authors.

Pre-registered on:

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1) Have any data been collected for this study already?

No, no data have been collected for this study yet.

2) What's the main question being asked or hypothesis being tested in this study?

A lot of work in recent 30 years has shown that the fusiform face area (FFA) in the human brain is a functionally specialized region for the perception of faces (Kanwisher et al. 1997; Wada and Yamamoto 2001; Kanwisher and Yovel 2006). In functional magnetic resonance imaging (fMRI) studies, FFA responds more strongly to human faces when compared to other visual stimulus categories, including non-human faces, bodies, objects, and scenes (Berman et al. 2010)

The primary object we aim to study is the evidential value of statistically significant focal tests of FFA face-selectivity in the published fMRI literature. Using Z-curve analysis, we will test whether the distribution of statistically significant tests for the Faces versus Non-Faces contrast in the fusiform face area (FFA) reflects genuine evidential value rather than selective reporting or statistical noise.

We predict that:

The Estimated Discovery Rate (EDR) will exceed the nominal alpha level (5%), and

The Estimated Replication Rate (ERR) will be substantially above zero

Evidence of low EDR (approximately 5%) combined with low ERR would indicate that the significant literature on FFA face-selectivity lacks robust evidential value.

3) Describe the key dependent variable(s) specifying how they will be measured.

The dependent variable for this analysis is the inferential test statistic corresponding to the confirmatory Faces versus Non-Faces contrast in the fusiform face area (FFA) from each eligible independent sample.

For each study, we will extract one focal test statistic (e.g., t-value, F-value, χ^2 statistic, correlation coefficient r , Z statistic, or exact p-value) that directly tests the Faces versus Non-Faces contrast in the FFA. Only inferential statistics will be extracted. Descriptive measures (e.g., percent BOLD signal change, GLM beta coefficients, or mean activation values) will not be treated as dependent variables unless they are accompanied by a reported inferential statistical test.

All extracted statistics will be converted into two-sided p-values, which will consequently be converted into absolute z-values. If a one-tailed p-value is explicitly reported, it will be multiplied by two to obtain a two-sided p-value. If only a test statistic is reported, the p-value will be computed using the appropriate distribution and degrees of freedom as specified in the Analysis section. The resulting absolute z-values (one per independent sample) will serve as the sole input to the Z-curve analysis.

4) How many and which conditions will participants be assigned to?

Because this is a meta-scientific analysis of published literature, no new participants will be assigned to experimental conditions. Instead, we will extract statistical tests corresponding to a predefined confirmatory contrast from eligible fMRI studies.

Eligible contrasts must test the activation difference between human faces versus non-face visual object stimuli (i.e., houses, tools, everyday objects, scenes, or scrambled images).

To qualify, the contrast must:

Explicitly compare human faces to a non-face visual category within the same experimental design.

Be described as a confirmatory test of face selectivity (not explicitly exploratory).

Target the fusiform face area (FFA), defined either functionally or anatomically.

The following contrasts will be excluded:

Faces versus fixation or baseline alone (without a non-face object comparison).

Faces versus inverted faces only.

Faces versus bodies only.

Contrasts described explicitly as exploratory post hoc analyses.

If multiple eligible Faces versus Non-Face contrasts are reported within the same independent sample, we will extract the contrast reported as the primary confirmatory result in the main results section.

5) Specify exactly which analyses you will conduct to examine the main question/hypothesis.

We will conduct a Z-curve analysis to evaluate the evidential value and expected replicability of statistically significant focal tests of the confirmatory Faces versus Non-Faces contrast in the fusiform face area (FFA).

The unit of analysis is one statistically significant focal test per independent sample. If multiple eligible contrasts are reported within the same independent sample, we will extract the primary confirmatory Faces versus Non-Faces contrast reported in the main results section.

Eligible reported statistics include exact p-values, t , F , χ^2 , r , or Z statistics. All statistics will be converted to two-sided p-values as follows:

Exact two-sided p-values will be recorded directly.

Explicit one-tailed p-values will be multiplied by 2.

t : $p = 2 \cdot \text{pt}(-|t|, df)$

r : convert to t using $t = r\sqrt{(n-2)/(1-r^2)}$, $df = n-2$

F : $p = \text{pf}(F, df1, df2, \text{lower.tail} = \text{FALSE})$

χ^2 : $p = \text{pchisq}(\chi^2, \text{df}, \text{lower.tail} = \text{FALSE})$

Z: $p = 2 \cdot \text{pnorm}(-|Z|)$

For inequality p-values (e.g., $p < .05$), p will be conservatively set to the reported alpha threshold before conversion. Only statistically significant results (two-sided $p \leq .05$) will be included. All p-values will be converted to absolute z-values using: $|z| = \text{qnorm}(1 - p/2)$

The resulting vector of $|z|$ values (one per independent sample) will be analyzed using the R package zcurve. We will report Estimated Discovery Rate (EDR), Estimated Replication Rate (ERR), and Expected False Discovery Rate (EFDR), along with 95% confidence intervals. Visualization will be produced using: `plot(fit, CI = TRUE, annotation = FALSE)`

Data extraction will be conducted independently by two coders using a shared extraction template. Discrepancies will be resolved by consensus or third-party adjudication.

6) Describe exactly how outliers will be defined and handled, and your precise rule(s) for excluding observations.

A study will be excluded if:

No eligible confirmatory contrast: The study does not report a confirmatory Faces vs Non-Faces contrast targeting the FFA.

No extractable test statistic: Only thresholded statistical maps are reported, or results are described without extractable inferential statistics.

Sample size unavailable: Sample size is not reported and cannot be inferred from the paper (or from reported degrees of freedom when inference is unambiguous).

Explicitly exploratory analyses only: The reported contrast is explicitly exploratory (e.g., described as exploratory post hoc without a confirmatory hypothesis/plan).

Reported statistic problems: if the reported statistic cannot be converted to a two-sided p-value; if the test reflects a manipulation check or secondary analysis rather than the primary confirmatory contrast; if the result duplicates a test from the same independent dataset.

A study will be included if satisfying the following criteria:

We will first systematically search the following journals but not limit them to these: Nature, Nature Neuroscience, Cell, Proceedings of the National Academy of Sciences (PNAS), PLOS Computational Biology, Journal of Neuroscience, NeuroImage.

Human fMRI studies in recent 35 years.

Containing Face versus Non-faces contrast in FFA, which can be defined functionally or anatomically.

Including statistical tests that reported:

exact p-value

or enough descriptive statistics to estimate p-value (e.g. F-statistic and degrees of freedom).

If p-values were reported for both left hemisphere and right hemisphere, choose the average bilateral p-values. If multiple confirmatory comparisons are performed and their p-values reported, we will take the most central, first-reported confirmatory result.

7) How many observations will be collected or what will determine sample size? No need to justify decision, but be precise about exactly how the number will be determined.

Our sample size is defined as the total number of focal statistical tests (z-scores) extracted from the eligible literature.

We aim to collect 25-50 focal tests to ensure a reasonably stable Z-curve estimate. However, the final sample size will be determined by the total number of papers that meet our explicit inclusion and exclusion criteria (see section 6).

8) Anything else you would like to pre-register? (e.g., secondary analyses, variables collected for exploratory purposes, unusual analyses planned?)

None